

Determinant of 3x3 matrices

Basic skills

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Q1. a) $\begin{vmatrix} -3 & 2 \\ -5 & 1 \end{vmatrix} = -3 \times 1 - 2 \times -5 = -3 + 10 = 7$

b) $\begin{vmatrix} 2 & 4 \\ -1 & -2 \end{vmatrix} = -4 - -4 = 0$

c) $\begin{vmatrix} -5 & 2 \\ 3 & 0.5 \end{vmatrix} = -2.5 - 6 = -8.5$

Q2. $M = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 5 \\ 0 & 3 & -2 \end{pmatrix}$

a) For m_{13} remove row 1, col 3 $\begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 5 \\ 0 & 3 & -2 \end{pmatrix}$
so $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$

b) For m_{22} remove row 2, col 2 $\begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 5 \\ 0 & 3 & -2 \end{pmatrix}$
so $\begin{pmatrix} 3 & 1 \\ 0 & -2 \end{pmatrix}$

Q3.

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

$$a_{ij} = (-1)^{i+j}$$

Q4.

$$\begin{vmatrix} 2 & -1 & 0 \\ 0 & -2 & 3 \\ 5 & 4 & 2 \end{vmatrix} = 2 \begin{vmatrix} -2 & 3 \\ 4 & 2 \end{vmatrix} - -1 \begin{vmatrix} 0 & 3 \\ 5 & 2 \end{vmatrix} \\ = 2(-4 - 12) + 1(0 - 15) \\ = 2 \times -16 - 15 = -47$$

Competency skills

Q5.

$$|M| = 3 \begin{vmatrix} 2 & 5 \\ 0 & -2 \end{vmatrix} - \begin{vmatrix} 2 & 1 \\ 0 & -2 \end{vmatrix} + 2 \begin{vmatrix} 2 & 1 \\ 2 & 5 \end{vmatrix}$$

by expanding on first row with minors

$$m_{11} = \begin{pmatrix} 2 & 5 \\ 0 & -2 \end{pmatrix} \quad m_{21} = \begin{pmatrix} 2 & 1 \\ 0 & -2 \end{pmatrix} \quad m_{31} = \begin{pmatrix} 2 & 1 \\ 2 & 5 \end{pmatrix}$$

$$\text{so } |M| = 3(-4 - 0) - (-4 - 0) + 2(10 - 2) \\ = -12 + 4 + 2 \times 8 = 8$$

Q6. a) $\begin{vmatrix} 5 & 3 & -2 \\ 2 & 0 & 4 \\ -1 & 1 & 2 \end{vmatrix} = 5 \begin{vmatrix} 0 & 4 \\ 1 & 2 \end{vmatrix} - 3 \begin{vmatrix} 2 & 4 \\ -1 & 2 \end{vmatrix} + -2 \begin{vmatrix} 2 & 0 \\ -1 & 1 \end{vmatrix}$

$$= 5(0 - 4) - 3(4 - -4) - 2(2 - 0)$$

$$= -20 - 4 = -24$$

b) $\begin{vmatrix} 6 & -4 & 1 \\ 0 & 2 & 3 \\ 5 & -3 & 5 \end{vmatrix} = 6 \begin{vmatrix} 2 & 3 \\ -3 & 5 \end{vmatrix} - -4 \begin{vmatrix} 0 & 3 \\ 5 & 5 \end{vmatrix} + 1 \begin{vmatrix} 0 & 2 \\ 5 & -3 \end{vmatrix}$

$$= 6(10 - -9) + 4(0 - 15) + (0 - 10)$$

$$= 6 \times 19 - 60 - 10 = 44$$

c) $\begin{vmatrix} -2 & -3 & 4 \\ 8 & -7 & -2 \\ 10 & 3 & -3 \end{vmatrix} = -2 \begin{vmatrix} -7 & -2 \\ 3 & -3 \end{vmatrix} - -3 \begin{vmatrix} 8 & -2 \\ 10 & -3 \end{vmatrix} + 4 \begin{vmatrix} 8 & -7 \\ 10 & 3 \end{vmatrix}$

$$= -2(21 - 6) + 3(-24 - -20) + 4(24 - -70)$$

$$= -2 \times 15 + 3 \times -4 + 4 \times 94 = 334$$

Q7. $\begin{vmatrix} \lambda & 1 & 3 \\ -1 & \lambda & 2 \\ 4 & 3 & \lambda \end{vmatrix} = \lambda \begin{vmatrix} \lambda & 2 \\ 3 & \lambda \end{vmatrix} - 1 \begin{vmatrix} -1 & 2 \\ 4 & \lambda \end{vmatrix} + 3 \begin{vmatrix} -1 & \lambda \\ 4 & 3 \end{vmatrix}$

$$\begin{aligned}
 &= \lambda (\lambda^2 - 6) - (-\lambda - 8) + 3(-3 - 4\lambda) \\
 &= \lambda^3 - 6\lambda + \lambda + 8 - 9 - 12\lambda \\
 &= \lambda^3 - 17\lambda - 1
 \end{aligned}$$

$$\text{so } \lambda^3 - 17\lambda - 1 = 3 \Rightarrow \lambda^3 - 17\lambda - 4 = 0$$

As $\lambda = -4$ solves $(\lambda + 4)$ is factor

$$(\lambda + 4)(\lambda^2 - 4\lambda - 1) = 0$$

$$\Rightarrow \lambda^2 - 4\lambda - 1 = 0 \Rightarrow \lambda = \frac{4 \pm \sqrt{16 + 4}}{2} = 2 \pm \sqrt{5}$$

Advanced skills

Q 8. It is the scale factor of volume change under a transform

Q9. $\begin{vmatrix} x & 1 & 1 \\ 0 & -x & 3 \\ 2x & 2 & 1 \end{vmatrix} \xrightarrow{\text{Col expand}} = x \begin{vmatrix} -x & 3 \\ 2 & 1 \end{vmatrix} - 0 + 2x \begin{vmatrix} 1 & 1 \\ -x & 3 \end{vmatrix}$

$$= x(-x - 6) + 2x(3 + x)$$

$$= -x^2 - 6x + 6x + 2x^2 = x^2$$

$$\begin{matrix} \frac{1}{2} & \frac{1}{2} \\ -2 & 2 \end{matrix}$$

for $x^2 \geq 4 \Rightarrow x^2 - 4 \geq 0$ so $x \geq 2$ or $x \leq -2$

Q10. Take

$$M = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

a) $M' = \begin{pmatrix} ka & kb & kc \\ d & e & f \\ g & h & i \end{pmatrix}$

$$\begin{aligned} \text{Then } |M'| &= ka \begin{vmatrix} e & f \\ h & i \end{vmatrix} - kb \begin{vmatrix} d & f \\ g & i \end{vmatrix} + kc \begin{vmatrix} d & e \\ g & h \end{vmatrix} \\ &= k \left(a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix} \right) \\ &= k |M| \end{aligned}$$

so multiplies determinant by constant

b) As can expand on any row/column, exchanging will have no affect on determinant

c) As kM is multiplying 3 rows by k by our result in a) $|kM| = k^3 |M|$

as need to multiply by k three times